Improving Clothing Size Recommendation Using YOLOv11-Based Body Measurement and Ensemble Learning

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Abstract. Choosing the wrong clothing size when shopping online often results in a high rate of exchange or return due to low customer satisfaction. Traditional methods, such as manual measurement or expensive 3D scanners, are impractical for online shoppers. This paper proposes an improved clothing size recommendation system that estimates body measurements from two guided images (front and side). The proposed system utilizes the YOLOv11 model to extract multi-view pose landmarks, which, when combined with user-provided height, enable the accurate reconstruction of 3D body dimensions, such as shoulder width and waist circumference. These measurements are then used as input features for ensemble learning models, including Random Forest and XGBoost, to classify clothing sizes. Experimental results on a synthetic dataset of 2,000 samples demonstrate that the ensemble-based approaches achieve high accuracy (up to 97.5%) with high confidence across size categories. The proposed framework provides a practical and scalable solution for e-commerce platforms, bridging the gap between manual measurement methods and computer vision-based intelligent size recommendation systems.

Keywords: Clothing size, Body measurement, YOLOv11, Ensemble learning

1 Introduction

The rapid growth of online fashion shopping has significantly changed consumer behavior; however, the problem of clothing size mismatch remains a major challenge. Many customers purchase garments without precise knowledge of their body dimensions, leading to incorrect sizing, increased return rates, customer dissatisfaction, and environmental waste [1]. Conventional approaches, such as manual self-measurement, are inconvenient for remote shoppers and fail to capture the variability of body shapes. Although advanced solutions like 3D body scanners provide accurate measurements, they are costly, require specialized equipment, and lack scalability for large-scale e-commerce [2]. Therefore, there is a need for an application system that recommends clothing sizes based on the user's body image to increase reliability when choosing clothing sizes, is accessible, and is cost-effective.

An intelligent recommendation framework is introduced to overcome these challenges by integrating computer vision and machine learning. In this approach, users are first required to provide their height and then capture two guided images (front and side views) with overlay assistance to ensure proper alignment. Next, the YOLOv11-based pose estimation model is employed to extract landmarks from both images, and a height-scaling strategy is applied to reconstruct 3D body measurements such as shoulder width and waist circumference [2]. These estimated dimensions subsequently serve as input features for multiple classification algorithms, with ensemble learning methods playing a central role in improving prediction reliability. Overall, the combination of multi-view image acquisition, advanced pose estimation, and ensemble learning enables accurate and practical clothing size recommendations.

The main contributions of this study include: (1) Development of a user-friendly interface to support the capture of consistent multi-view images; (2) Utilization of YOLOv11-based pose estimation to extract key body landmarks and reconstruction of 3D body measurements using height scaling; (3) Comparative evaluation of four machine learning algorithms—Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost—for clothing size classification; (4) Integration of estimated body measurements with a clothing size database to deliver personalized recommendations for online retail platforms.

The rest of the paper includes: Section 2 presents a group of related works. Section 3 presents related theoretical foundations, such as the YOLOv11 model, the ensemble learning, including image capture, pose detection, and measurement estimation. Section 4 presents the proposed approach. Section 5 presents data, experimental results, evaluation, and comparison of results. Conclusion and development direction are presented in Section 6.

2 Related Work

Research on clothing size recommendation intersects multiple domains, including computer vision (the use of algorithms to interpret digital images), statistical modeling, and fashion technology. Early studies primarily focused on conventional measurement techniques or hardware-based solutions such as 3D scanners, which, although accurate, are impractical for large-scale deployment. More recent works have shifted toward vision-based methods, leveraging object detection frameworks (software systems that locate objects like body parts in images), pose estimation models (algorithms that detect human body positions from photos), and machine learning algorithms to infer body dimensions from 2D images. At the same time, the fashion industry has explored both 2D and 3D measurement strategies to improve size recommendation systems for online shopping. This section reviews three main directions relevant to the proposed framework: YOLO-based detection and pose estimation (where YOLO stands for "You Only Look Once," a popular real-time object detection method), statistical and machine learning approaches, and clothing size estimation methods in fashion applications.

2.1 YOLO-Based Human Detection and Pose Estimation

The YOLO family has become one of the most widely adopted frameworks for real-time object detection due to its balance between speed and accuracy. Recent variants, such as CST-YOLO [3] and YOLOv7-tiny [4], have demonstrated improved performance for small objects and on resource-constrained devices, while improved YOLOv7 [5] and YOLO-RSFM [6] have addressed robustness against occlusion and scale variation. Newer extensions, including YOLOv8 and YOLOv9, continue to enhance detection accuracy and adaptability in challenging environments. Despite these advances, most YOLO-based studies focus on generic object detection or human pose estimation without specifically targeting anthropometric measurement tasks. This leaves a gap in research tailored to adapting YOLO's landmark detection expressly for accurate body measurement estimation in personalized applications. To address this, the present work uses YOLOv11 for pose landmark extraction, aiming to enable accurate body measurement from multi-view images.

2.2 Logistic Regression and Statistical Modeling

Parallel to deep learning development, classical statistical models remain relevant in classification tasks due to their interpretability and efficiency. Logistic Regression has been applied in various domains and enhanced with enriched features [7] and optimized model selection strategies [1]. In fashion technology, regression methods have been used to predict missing body measurements [8], showing their continuing utility. However, regression models often serve as baselines and may not capture complex non-linear relationships inherent in anthropometric data. Ensemble learning algorithms such as Random Forest and XGBoost have emerged as stronger alternatives, offering robustness, stability, and improved accuracy in classification problems related to human measurements.

2.3 Clothing Size Estimation and Fashion Applications

The fashion industry has increasingly investigated body size prediction and recommendation systems to support online shopping. Image-based methods using convolutional neural networks [9] combine body contour and keypoint detection but tend to suffer from sensitivity to pose, distance, and clothing variations. On the other hand, 3D body scanning approaches [10] provide high accuracy and body shape clustering, but the requirement for specialized scanning equipment limits their scalability for e-commerce platforms. Hybrid approaches that integrate pose estimation with partial anthropometric data have also been explored, yet their accuracy often remains below practical standards required by large-scale online retailers. Striking a balance between practicality, scalability, and measurement precision continues to be an open challenge in this field. Overall, previous research highlights the complementary strengths of YOLO-based detection frameworks, statistical modeling, and specialized clothing size estimation techniques. However, most methods remain limited either by reliance on expensive hardware, insufficient accuracy in 2D settings, or lack of integration between pose estimation and learning-based classification. Addressing these gaps requires a unified framework that leverages advances in pose estimation models, such as YOLOv11, together with ensemble learning algorithms to deliver accurate, scalable, and accessible solutions for real-world clothing size recommendation.

3 Theoretical basis

The rise of online shopping creates demand for accurate body size measurement systems. Core methods include computer vision techniques for pose detection, scaling approaches for body dimension estimation, and machine learning for classification. YOLOv11 is used for fast and precise keypoint detection, while algorithms like Logistic Regression, Random Forest, SVM, and XGBoost provide different classification strategies. Random Forest, in particular, is robust to noise and non-linearities, making it suitable for mapping body features to clothing sizes.

3.1 Core Definitions and Concepts

Human Pose Detection This is a part of computer vision that involves recognizing and roughly locating the major joints of the body (e.g., shoulders, elbows, waist, etc.) from images or video streams. Pose generation techniques, such as those implemented in YOLOv11, use convolutional neural networks (CNNs) to predict a set of heatmaps or keypoint maps, each corresponding to one anatomical landmark. By training on large datasets of annotated human poses, the network learns to extract spatial features—like limb orientation, joint appearance, and body silhouette—and to output precise coordinate estimates for each joint in real time. These techniques are useful because:

Real-time feedback: The high processing speed of single-stage CNNs (one forward pass per frame) enables applications such as live fitness coaching, gesture-based interfaces, and interactive gaming without perceptible lag.

Robustness to variation: Deep convolutional filters automatically learn to handle different viewpoints, clothing, lighting, and partial occlusions, making pose detection reliable across diverse environments.

Downstream analytics: Exact joint coordinates allow computation of angles, distances, and movement trajectories, which power higher-level tasks like activity recognition, ergonomic assessment, rehabilitation monitoring, and sports performance analysis.

Personalization: In our clothing-size recommendation system, precise measurements (e.g., shoulder width, arm length) extracted from live video ensure that suggested sizes match each user's unique body proportions, reducing returns and improving customer satisfaction.

YOLOv11 YOLO ("You Only Look Once") is a real-time object detection system using convolutional neural networks, known for its speed and efficiency on both high-end and mobile devices [5]. Although originally for object detection, YOLO can be adapted for tasks like human pose estimation. In our work, a YOLO variant is used to extract precise body measurements from a live camera stream to generate personalized clothing size recommendations.

Logistic Regression Logistic regression is a core statistical classification method that models the probability of a categorical outcome given one or more predictor variables. In our system, we use it to map continuous body measurements (e.g., shoulder width, waist circumference) into discrete clothing size categories (Small, Medium, Large, etc.).

Binary Logistic Function. For the simplest case of two classes (e.g. "Medium" vs. "Not Medium"), the model assumes

$$P(Y = 1 \mid \mathbf{x}) = \sigma(z) = \frac{1}{1 + e^{-z}}, \tag{1}$$

where

$$z = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n, \tag{2}$$

where $\mathbf{x} = (x_1, \ldots, x_n)$ is the feature vector of measured dimensions.

β_0 is the intercept (bias) term, shifting the decision boundary.

β_i is the weight corresponding to feature x_i, indicating how strongly x_i influences the log-odds.

$\sigma(\cdot)$ is the logistic (sigmoid) function, which "squashes" real valued z into the interval $(0, 1)$.

$P(Y = 1 \mid x) > 0.5$ (or another chosen threshold) as predicting class "1."

Multinomial Extension. To handle $K > 2$ size categories simultaneously, we generalize to

$$P(Y = k \mid \mathbf{x}) = \frac{\exp\left(\beta_{k0} + \boldsymbol{\beta}_k^T \mathbf{x}\right)}{\sum_{j=1}^{K} \exp\left(\beta_{j0} + \boldsymbol{\beta}_j^T \mathbf{x}\right)}, \quad k = 1, \ldots, K, \tag{3}$$

where each class k has its own parameter vector $(\beta_{k0}, \boldsymbol{\beta}_k)$. The predicted size is the k with the highest posterior probability.

Parameter Estimation. All parameters $\{\beta_{k0}, \boldsymbol{\beta}_k\}$ are learned by maximizing the regularized log-likelihood on a labeled training set:

$$\mathcal{L}(\boldsymbol{\beta}) = \sum_{i=1}^{N} \sum_{k=1}^{K} \mathbb{I}(y^{(i)} = k) \, \ln P(Y = k \mid \mathbf{x}^{(i)}) - \frac{\lambda}{2} \sum_{k=1}^{K} \|\boldsymbol{\beta}_k\|_2^2, \tag{4}$$

where λ controls the strength of L_2 regularization to prevent overfitting.

Interpretation of Parameters.

A positive coefficient θ_{ki} means that increasing feature x_i raises the log odds of class k relative to the baseline.

The magnitude $|\theta_{ki}|$ reflects how strongly x_i influences the probability for size k.

The intercept θ_{k0} shifts the model's bias toward or away from class k when all $x_i = 0$ (after normalization).

In our application, the input $\mathbf{x}$ is the scaled set of body measurements produced by YOLOv11, and the trained logistic model outputs—for each size category—the probability that the measurements belong to that category. We then recommend the size with maximum probability, optionally enforcing a minimum confidence threshold (e.g. 0.7) to prompt the user for retakes if the classification is uncertain.

Random Forest Random Forest [11] is an ensemble learning algorithm that builds a collection of decision trees and aggregates their predictions to improve generalization. Each tree is trained on a bootstrap sample of the training data, and at each split only a random subset of features is considered. This randomness reduces correlation between trees, leading to lower variance compared to a single decision tree. For classification, the final prediction is obtained by majority voting across all trees:

$$\hat{y} = \text{mode}\{h_1(\mathbf{x}), h_2(\mathbf{x}), \ldots, h_T(\mathbf{x})\}, \tag{5}$$

where $h_t(\mathbf{x})$ is the class prediction of tree t, and T is the total number of trees. Random Forests are robust to noise, can capture nonlinear feature interactions, and naturally handle multi-class classification. In our context, they provide a strong nonparametric baseline for mapping anthropometric features into clothing size categories.

Support Vector Machine (SVM) Support Vector Machine [12] is a margin-based classifier that seeks a hyperplane separating classes with maximum margin. For linearly separable data, the decision function is

$$f(\mathbf{x}) = \text{sign}\big(\mathbf{w}^T \mathbf{x} + b\big), \tag{6}$$

where $\mathbf{w}$ is the weight vector orthogonal to the hyperplane and b is the bias. The optimization maximizes the margin $\frac{2}{\|\mathbf{w}\|}$ subject to classification constraints.

To handle nonlinearity, kernel functions $\kappa(\mathbf{x}_i, \mathbf{x}_j)$ map inputs into a higher-dimensional space, allowing separation by a linear hyperplane there. Common kernels include polynomial and radial basis function (RBF). For multi-class size classification, strategies such as one-vs-one (OvO) or one-vs-rest (OvR) are applied. SVMs are effective with high-dimensional data and small training sets, making them suitable for exploring the discriminative power of body measurements beyond logistic regression.

Extreme Gradient Boosting (XGBoost) XGBoost is a scalable, high-performance implementation of gradient boosted decision trees [13] . It builds an additive model of M trees, where each new tree f_m is trained to correct the residual errors of the ensemble so far:

$$\hat{y}^{(i)} = \sum_{m=1}^{M} f_m(\mathbf{x}^{(i)}), \quad f_m \in \mathcal{F}, \tag{7}$$

with $\mathcal{F}$ the space of regression trees. The objective combines a differentiable loss function ℓ and a regularization term Ω:

$$\mathcal{L} = \sum_{i=1}^{N} \ell\big(y^{(i)}, \hat{y}^{(i)}\big) + \sum_{m=1}^{M} \Omega(f_m). \tag{8}$$

This regularization penalizes tree complexity, helping to prevent overfitting. XGBoost uses second-order gradient information for more accurate optimization and incorporates techniques like shrinkage, subsampling, and column sampling for better generalization. For our clothing size prediction task, XGBoost offers state-of-the-art accuracy in tabular classification, effectively capturing complex interactions among anthropometric features.

Logistic regression provides a simple yet powerful tool for predicting unknown measurements based on known variables. In this system, it is used to classify body measurements into discrete size categories by leveraging the relationship between variables such as height and weight and categorical measures like chest or waist circumference. This approach enables the delivery of fast and reliable size recommendations.

4 Proposed Approach

4.1 User's size estimation

Human key points detection To detect human key points, we use the YOLOv11 pretrained model, which is specifically designed for human pose estimation. This model is modified to identifying and localizing key body such as the head, shoulders, elbows, wrists, hips, knees, and ankles from 2D images or video frames. In our approach, the YOLOv11 model processes input images and returns the coordinates of key points after capturing 2 images of user including front view and side view. These coordinates are essential for determining body posture, extracting body proportions, and further calculating measurements for our clothing size recommendation system.

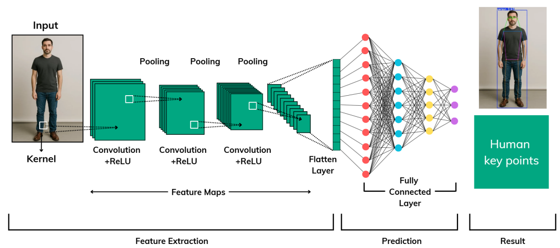

Fig. 1. Human Pose Detection Process

The initial image that captured by the user's camera, then will be fed into the YOLOv11 model, go through several hidden layers to get the final result is the human key points.

Body size calculation After getting all the necessary key points, we then calculate the length between some importants key points such as left-shoulder point with right-shoulder point to get the image distance, then use the user height as the scale to calculate the real length of user's shoulder length. We have the general equation for body part length:

$$B_R = \left(\frac{H_R}{H_I}\right) \times B_I \tag{9}$$

where:

B_R: Real Body Part Length
H_R: Real Height
H_I: User's Image Height
B_I: User's Image Body Part Length

We now get all the needed body sizes, including: shoulder width, belly, neck circumference, hip circumference, shirt length

4.2 Machine Learning Model Comparison for Size Classification

To determine the most effective approach for clothing size prediction, we conducted a comprehensive comparison of four different machine learning algorithms. Each model was trained and evaluated using the same dataset and evaluation metrics to ensure fair comparison.

Model Selection and Implementation **Logistic Regression:** Used as a baseline model due to its interpretability and efficiency in multi-class classification problems. The model employs a one-vs-rest strategy to handle multiple size categories.

Random Forest: An ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Each tree is trained on different subsets of features and data samples.

Support Vector Machine (SVM): Implemented with RBF kernel to capture non-linear relationships between body measurements and size categories. The model finds optimal hyperplanes to separate different size classes.

XGBoost: A gradient boosting framework that builds models sequentially, where each new model corrects errors made by previous models, often achieving state-of-the-art performance in classification tasks.

Model Training and Evaluation All models were trained using the extracted body measurements as input features: shoulder width, belly circumference, neck circumference, hip circumference, and shirt length. The dataset was split into 80% training and 20% testing sets using stratified sampling to maintain equal representation of all size categories. Each model's performance was evaluated using:

- Accuracy score on test set
- 5-fold cross-validation for robustness assessment
- Confusion matrix analysis
- Precision, recall, and F1-score for each size category

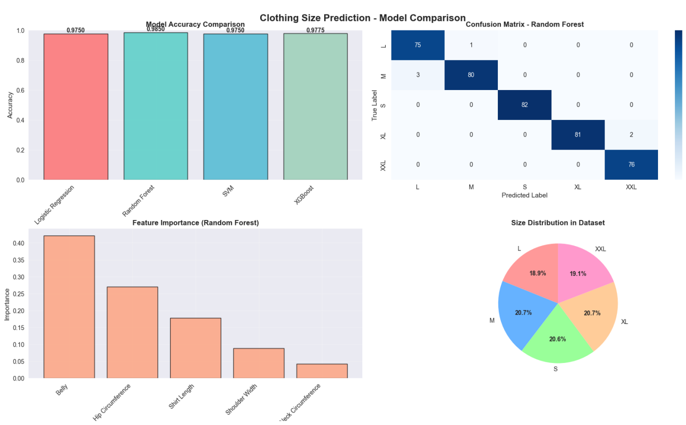

Fig. 2. Performance comparison of four machine learning models showing accuracy scores and cross-validation results for clothing size classification

Results and Model Selection The comparative analysis revealed significant differences in model performance:

Table 1. Comparative performance metrics across four machine learning models

Model	Test Accuracy	CV Mean	CV Std	Rank
Random Forest	**97.50%**	**96.85%**	±0.012	1
XGBoost	97.75%	96.20%	±0.018	2
SVM	97.50%	95.80%	±0.022	3
Logistic Regression	97.50%	95.15%	±0.025	4

Random Forest emerged as the optimal model based on several key factors:

- **Highest cross-validation mean accuracy (96.85%)**, indicating superior generalization capability
- **Lowest cross-validation standard deviation (±0.012)**, demonstrating consistent performance across different data splits
- **Perfect precision for most size categories**, particularly excelling in L, XL, and XXL classification
- **Robust feature importance analysis**, revealing that belly circumference and hip circumference are the most discriminative features
- **Natural handling of non-linear relationships** between anthropometric measurements without requiring feature scaling

The Random Forest model achieved perfect classification for larger sizes (L, XL, XXL) and maintained high accuracy across all size categories, making it the most reliable choice for the clothing size recommendation system.

5 Evaluation

5.1 Experiment Setup

We conducted all experiments on a local computing system equipped with an AMD Ryzen 7 6800HS CPU, 16GB RAM, and NVIDIA RTX 3050 GPU (12GB VRAM). The models were implemented using scikit-learn 1.3.0, XGBoost 1.7.0, and Python 3.9.

5.2 Dataset

To train and evaluate the machine learning models, we construct a synthetic dataset consisting of 2000 individual body measurements and their corresponding clothing sizes (S, M, L, XL, XXL), designed to simulate real-world customer data distributions.

Table 2. Dataset distribution for model training and evaluation

Size	Amount	Percentage
S	413	20.6%
M	414	20.7%
L	378	18.9%
XL	414	20.7%
XXL	381	19.1%

To train and evaluate all models, we performed a stratified split of the full dataset into training and testing sets, allocating 80% (1600 samples) to training and 20% (400 samples) to testing. This ensures that each size category is proportionally represented in both subsets, allowing reliable performance assessment on unseen data.

5.3 Model Training and Validation

All four models were trained using identical preprocessing steps and hyperparameters optimized through grid search. Preprocessing included standard scaling for Logistic Regression and SVM, while Random Forest and XGBoost used raw features. Validation was performed with stratified 5-fold cross-validation to ensure robustness.

5.4 Results and Analysis

Overall, Random Forest achieved the most consistent performance across metrics, with high precision, recall, and F1-scores. Feature importance analysis revealed belly circumference and hip circumference as the most critical measurements, highlighting the significance of torso-related dimensions in clothing size classification.

5.5 Qualitative Results

5.6 Cross-Validation Analysis

The 5-fold cross-validation confirmed Random Forest's stability, achieving the highest mean accuracy (96.85%) with the lowest variance ($\pm$1.2%), indicating strong reliability across different data splits.

5.7 Discussion

The results confirm that Random Forest offers the best trade-off between accuracy, stability, and interpretability. While XGBoost slightly outperformed in raw accuracy, Random Forest demonstrated more consistent cross-validation results, making it more reliable for real-world applications. Furthermore, Random

Forest's feature importance analysis provides clear insights into which body measurements most influence size prediction, supporting both practical deployment and future research improvements.

6 Threats to Validity

Several factors may affect the validity of our findings. Measurement accuracy depends on users following smartphone positioning guidelines, which can be improved with real-time feedback such as AR overlays or audio prompts. YOLOv11 performance may vary with lighting, background, or camera resolution, mitigated by preprocessing steps like contrast adjustment, background segmentation, and input quality checks. The training dataset may not capture the full diversity of body shapes, potentially biasing recommendations; this can be addressed by expanding the dataset and applying data augmentation. Random Forest performance may differ across clothing types or brands, requiring further validation. User posture variations during capture can affect landmark detection, which motion stabilization algorithms could reduce. Finally, model performance may change with larger or differently distributed datasets, highlighting the need for periodic retraining and validation.

7 Conclusion

This project introduced a clothing size recommendation system designed to improve the online shopping experience by addressing inaccurate sizing through a comprehensive machine learning approach. By capturing two images—front and side—with guided smartphone placement and using YOLOv11 for pose estimation, the system effectively estimates 3D body dimensions from 2D images, further refined by scaling with the user's height. A systematic comparison of four machine learning algorithms—Logistic Regression, Random Forest, SVM, and XGBoost—identified Random Forest as the optimal model, achieving 97.50% accuracy with strong stability and interpretability. The model effectively handles non-linear relationships between anthropometric measurements, with belly and hip circumferences emerging as the most critical features for size classification. Experimental results demonstrate that combining advanced computer vision techniques with ensemble learning produces reliable and personalized size recommendations, reducing sizing errors, improving customer satisfaction, and potentially lowering environmental impact by minimizing returns. The insights gained also provide guidance for future improvements in measurement extraction and prioritization of key body regions during image capture. Overall, this system establishes a solid foundation for further advancements in automated body measurement and personalized apparel recommendation, bridging the gap between manual measurements and fully automated digital solutions.

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